

SOCIAL NETWORKS SEMINAR - Soc 63901

Instructors: Profs. David Hachen & Omar Lizardo

CONTACT INFORMATION

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SEMINAR OBJECTIVES

This seminar will examine both classical and more recent research on and methods for studying social networks.

The focus will be on both:

(a) Important substantive, theoretical and sociological issues that network analysis can address.

(b) The methods, techniques and statistical models for analyzing social networks. The specific topics that we will explore include: graph theory, ego networks, centrality, cliques, clustering, affiliation networks, structural equivalence and block modeling, local structure (dyads and triads), balance theory, homophily, reciprocity, tie strength, tie persistence, triadic configurations, longitudinal modeling, statistical models, small world networks, clustering, scale-free networks, diffusion, network science, and network effects.

SEMINAR REQUIREMENTS

This seminar will be a combination of lectures about and discussions of the course readings. There are three basic requirements:

1. **Research Paper:** A 15-20 publishable-like paper containing a review literature, hypotheses, data analysis and discussion of findings about a social network using some of the methods and ideas of social network analysis. Completing this paper

will require a series of steps (see Assignments & Calendar for due dates and more details):

- i. submitting a topic
 - ii. meeting with David & Omar to discuss the topic
 - iii. constructing an annotated bibliography of important literature on your topic
 - iv. describing possible data sources and detailing the pros and cons of each source
 - v. submitting a final research paper
2. **Weekly Homework Assignments:** These are designed to build familiarity with network methods, software, and analysis techniques. Assignments must be submitted, but they will be self graded and we will go over them in class.
3. **Class Participation:** Seminar participants are required to come prepared to seminar sessions by reading the required readings and preparing questions that they want discussed in class.

Final Grades for the course will be computed as follows:

Final Paper: 65%

Homework: 30%

Participation: 5%

SEMINAR READINGS

We have created an extensive reading list of both required and recommended readings (click on Reading List on home page for the pdf of this list).

Seminar participants are required to purchase the following books (they are available through the ND bookstore):

- Stanley Wasserman and Katherine Faust. 1997. *Social Network Analysis: Methods and Applications*. Cambridge: Cambridge University Press.
- Carrington, Peter J., John Scott, and Stanley Wasserman (eds.) 2005. *Models and Methods in Social Network Analysis*. Cambridge: Cambridge University Press.*
- Barabasi, Albert-Laszlo. 2003. *Linked*. New York, Plum (Penguin).

*The required readings from this book are also available online through RefWorks (see below). We still recommend purchasing this book because of the many interesting readings in it.

We have also ordered two other books which we recommend, but are not required:

- Degenne, Alain and Michel Forsé. 1999. *Introducing Social Networks*. Thousand Oaks, CA: Sage Publications.
- Burt, Ronald S. 1992. *Structural Holes: The Social Structure of Competition*. Cambridge, MA: Harvard University Press.

All other readings are available online through RefWorks, a web-based bibliographical database management system that all ND students can use. The citations for all required and recommended readings (and even some other stuff) have been placed in this database. For all required readings a pdf of the article is attached (thanks to Cheng Wang!). For all other readings that are available through an electronic journal, the "Find Text" button the corresponds to the citation will take you to pdf of the reading (as long as ND has an electronic subscription). You can also link to this database and use a cite-while-you-write tool to insert references in MS word documents.

To access the reading list database go to [RefWorks](#) and enter the following:

Log-in Name: socnet

Password: links

This will give you read only access to the database. Though you can not edit and add citations, you can download pdfs and use the citation tools.

Cheng has organized the readings by our weekly topics. Use the Folders pull-down menu to select a specific folder. Citations with a paper click icon have an attached pdf (these are usually required readings). Click on the "View" link for a specific citation to see the whole citation and a link to the pdf. Click on the "Find Text" button to find a pdf through NDs electronic journal subscriptions.

Note: Some of the readings will be math-heavy. However, being a math-wiz is not a requirement for getting a great deal out of this class. Our job will be in part to help you make sense of some of the more difficult readings. Our suggestion is for you to (1) get as much as you can get out of your first reading of a piece, (2) if you get stuck on a math-heavy section skim over it, look for a non-mathematical summary at the end of the section, and skip ahead until the article begins to make more sense, and (3) after we have gone over some of the the math in class, re-read the difficult sections.

SOFTWARE

Seminar participants should purchase [UCINET](#), a general network software package for Windows machines. A full operational trial version can be used for 60 days; student licences cost \$40. We are in the process of getting UCINET installed on some of the machines in the computer lab on the 9th floor of Flanner.

In all likelihood we will be using [PNet](#), a program for the estimation of p^* models. The software is free.

Other possibilities include:

- [StocNet](#) (Free download; fits a variety of models including p^*)
- [Pajek](#) (General suite; free for non-commercial use; does most of the basic stuff)
- [NETDRAW](#) (the network imaging program that accompanies UCINET, available for purchase if you want to make pretty pictures).

Other Software

- [ZO v. 2.3](#) Tom Snijder's simulation program to create a distribution of graphs with fixed characteristics (degree distribution, number of nodes, number of edges, etc.) for purposes of statistical inference.

Related Resources:

- Mo-Han Hsieh's [introduction to UCINET](#)
- Bob Hanneman and Mark Riddle's [UCINET tutorial](#).

Carter Butts, "[Social Network Analysis with sna](#)."